

For different values of λ ($0 \leq \lambda \leq 1$), we get an infinite number of optimal solutions.

Ex. 10. The given table represents a certain stage in the solution of an L. P. P. by the simplex method. Here M is a very large positive number and the problem is a maximization problem. State the original problem.

		c_j	3	-1	-1	0	0	$-M$	$-M$
c_B	B	b	a_1	a_2	a_3	a_4	a_5	a_6	a_7
0	a_4	10	3	-2	0	1	0	0	-1
$-M$	a_6	1	0	1	0	0	-1	1	-2
-1	a_3	1	-2	0	1	0	0	0	1
$z_j - c_j$		$-M-1$	-1	$-M+1$	0	0	M	0	$3M-1$

[Calcutta Hons., 1981]

The problem has three constraints. For its solution, one slack variable, one surplus variable and two artificial variables have been used.

It is seen from the given table that the present table has been obtained from the first table by replacing the artificial vector a_7 by the vector a_3 from the initial basis (a_4, a_6, a_7). a_6 and a_7 are artificial vectors while a_4 corresponds to the slack variable x_4 . Hence, if **B** be the present basis, then

$$B^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then computing $x_B = B^{-1}b$ and $Y_j = B^{-1}a_j$ where $j = 1, 2, \dots, 7$, we get **b** and $a_1, a_2, a_3, a_4, a_5, a_6, a_7$ of the previous table and they are

$$b = \begin{bmatrix} 11 \\ 3 \\ 1 \end{bmatrix}, a_1 = \begin{bmatrix} 1 \\ -4 \\ -2 \end{bmatrix}, a_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, a_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \dots$$

Also we see that $c_1 = 3$, $c_2 = -1$ and $c_3 = -1$.

The two artificial variables had been used, one for the constraint inequation of the greater than type and the other for the third constraint which is an equation.

Hence the original problem is

$$\text{Maximize } z = 3x_1 - x_2 - x_3$$

$$\text{subject to } x_1 - 2x_2 + x_3 \leq 11,$$

$$-4x_1 + x_2 + 2x_3 \geq 3,$$

$$-2x_1 + x_3 = 1,$$

$$x_1, x_2, x_3 \geq 0.$$

EXAMPLES V (A)

Q1 An intermediate tableau of an L. P. P. by simplex method is given in an incomplete form.

(i) Complete the row giving $(z_j - c_j)$ for all j .

(ii) Does this basic solution give the optimal value of the objective function of this problem?

04 Following is the starting tableau of an L. P. P. by the simplex method, in an incomplete form.

c_B	B	x_B	c_j	3	2	0	0	$-M$
			b	a_1	a_2	a_3	a_4	a_5
			2	2	1	1	0	0
			12	3	4	0	-1	1

- Complete the columns of c_B , B and x_B .
- Is there any vector corresponding to the slack and surplus variables?
- What is the object of introducing the vector a_5 ?
- Complete the row giving $(z_j - c_j)$.
- Write down the problem in its standard form.
- Write down the actual problem.
- Find the departing and the entering vectors and write down the next tableau.

[Ans (ii) a_3 (slack), a_4 (surplus). (iv) $(-3M - 3)$, $(-4M - 2)$, 0, M , 0.

(vi) Max $z = 3x_1 + 2x_2$
s. t. $2x_1 + x_2 \leq 2$, $3x_1 + 4x_2 \geq 12$ and $x_1, x_2 \geq 0$.]

05 The following incomplete tableau represents the second stage in the solution of an L. P. P. by the simplex method. All variables corresponding to zero co-efficients of the objective function are slack variables. Complete the tableau. Also state the original problem, assuming it to be a maximization problem. (The notations have their usual meanings).

$c_j \rightarrow$								0	0	0
c_B	Vectors in the basis	b	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8
0	a_6	9	13		1	-1	9			3
0	a_7	4	-2		0	4	1			1
	a_2	2	4		0	-1	1			1
$z_j - c_j$		8	15		2	-7	5			4

[Ans: $c_B = (0, 0, 4)$, $c = (1, 4, -2, 3, -1, 0, 0, 0)$.

$a_2 = (0, 0, 1)$, $a_6 = (1, 0, 0)$, $a_7 = (0, 1, 0)$.]

[Calcutta Hons., 1980]

Solve the following L. P. P. by simplex method (6 - 14) :

In each iteration, satisfy yourself that the vectors in the basis are linearly independent.

06 Maximize $z = 2x_1 + 3x_2$

subject to $x_1 + x_2 \leq 1$,

$3x_1 + x_2 \leq 4$,

$x_1, x_2 \geq 0$.

Check the result graphically.

[Ans: $x_1 = 0$, $x_2 = 1$; $z_{max} = 3$.]

07 Maximize $z = 5x_1 + 3x_2$
 subject to $3x_1 + 5x_2 \leq 15$,
 $5x_1 + 2x_2 \leq 10$,
 $x_1, x_2 \geq 0$.

[Visva-Bharati, 1983]

Check the result graphically.

[Ans. $x_1 = \frac{20}{19}$, $x_2 = \frac{45}{19}$, $z_{\max} = \frac{235}{19}$.]

08 Maximize $z = 4x_1 + 7x_2$
 subject to $2x_1 + x_2 \leq 1000$,
 $10x_1 + 10x_2 \leq 6000$,
 $2x_1 + 4x_2 \leq 2000$,
 $x_1, x_2 \geq 0$.

[Ans. $x_1 = 200$, $x_2 = 400$; $z_{\max} = 3600$.]

Check the result graphically.

09 Maximize $z = 3x_1 + x_2 + 3x_3$
 subject to $2x_1 + x_2 + x_3 \leq 2$,
 $x_1 + 2x_2 + 3x_3 \leq 5$,
 $2x_1 + 2x_2 + x_3 \leq 6$,
 $x_1, x_2, x_3 \geq 0$.

[Burdwan Hons., 1982]

[Ans. $x_1 = \frac{1}{5}$, $x_2 = 0$, $x_3 = \frac{8}{5}$; $z_{\max} = \frac{27}{5}$.]

10 Minimize $z = x_1 - 3x_2 + 2x_3$
 subject to $3x_1 - x_2 + 2x_3 \leq 7$,
 $-2x_1 + 4x_2 \leq 12$,
 $-4x_1 + 3x_2 + 8x_3 \leq 10$,
 $x_1, x_2, x_3 \geq 0$.

[North Bengal Hons., 2008]

[Ans. $x_1 = 4$, $x_2 = 5$, $x_3 = 0$; $z_{\min} = -11$.]

11 Maximize $z = 3x_1 + 2x_2 + 5x_3$
 subject to $x_1 + 2x_2 + x_3 \leq 430$,
 $3x_1 + 2x_3 \leq 460$,
 $x_1 + 4x_2 \leq 420$,
 $x_1, x_2, x_3 \geq 0$.

[Calcutta Hons., 1983]

[Ans. $x_1 = 0$, $x_2 = 100$, $x_3 = 230$; $z_{\max} = 1350$.]

12 Maximize $z = 2x_1 + 4x_2 + x_3 + x_4$
 subject to $x_1 + 3x_2 + x_4 \leq 4$,
 $2x_1 + x_2 \leq 3$,
 $x_2 + 4x_3 + x_4 \leq 3$,
 $x_1, x_2, x_3, x_4 \geq 0$.

Satisfy yourself that the matrix written in the final tableau in place of the initial basis matrix gives the inverse of the basis at the final iteration. [Delhi, 1977]

[Ans. $x_1 = 1$, $x_2 = 1$, $x_3 = \frac{1}{2}$, $x_4 = 0$; $z_{\max} = \frac{13}{2}$.]

13 Maximize $z = 3x_1 + 4x_2 + x_3 + 5x_4$
 subject to $8x_1 + 3x_2 + 2x_3 + 2x_4 \leq 10$,
 $2x_1 + 5x_2 + x_3 + 4x_4 \leq 5$,
 $x_1 + 2x_2 + 5x_3 + x_4 \leq 6$,
 $x_1, x_2, x_3, x_4 \geq 0$.

[Ans. $x_1 = \frac{15}{14}$, $x_2 = x_3 = 0$, $x_4 = \frac{5}{7}$; $z_{\max} = \frac{95}{14}$.]



- 14 Maximize $z = 4x_1 + 5x_2 + 9x_3 + 11x_4$
 subject to $x_1 + x_2 + x_3 + x_4 \leq 15$,
 $7x_1 + 5x_2 + 3x_3 + 2x_4 \leq 120$,
 $3x_1 + 5x_2 + 10x_3 + 15x_4 \leq 100$,
 $x_1, x_2, x_3, x_4 \geq 0$.

[Ans. $x_1 = \frac{50}{7}$, $x_2 = 0$, $x_3 = \frac{55}{7}$, $x_4 = 0$; $z_{max} = \frac{695}{7}$.]

- 15 Find $x_1, x_2, x_3 \geq 0$
 subject to $x_1 + x_2 - 2x_3 \leq 10$,
 $4x_1 + x_2 + x_3 \leq 20$,

[Kalyani Hons., 1988]

which will maximize $z = 10x_1 + x_2 + 2x_3$.

Satisfy yourself that the matrix written in each tableau in place of the initial basis matrix gives the inverse of the basis of that iteration. [Calcutta M. Sc., 1977]

[Ans. $x_1 = 5$, $x_2 = 0$, $x_3 = 0$; $z_{max} = 50$.]

- 16 Solve the following L. P. P. by any method :

- (i) Maximize $z = 7x_1 + 3x_2$, $x_1, x_2 \geq 0$
 subject to $x_1 + 2x_2 \geq 3$,
 $x_1 + x_2 \leq 4$,
 $x_1 \leq \frac{5}{2}$,
 $x_2 \leq \frac{3}{2}$.

[Jadavpur, 1984]

[Ans. $x_1 = \frac{5}{2}$, $x_2 = \frac{3}{2}$; $z_{max} = 22$.]

- (ii) Minimize $z = 4x_1 + 8x_2 + 3x_3$, $x_1, x_2, x_3 \geq 0$
 subject to $x_1 + x_2 \geq 2$,
 $2x_1 + x_3 \geq 5$.

[Calcutta Hons., 1990]

[Ans. $x_1 = \frac{5}{2}$, $x_2 = x_3 = 0$; $z_{min} = 10$.]

- 17 Solve the following L. P. P. by introducing artificial variables :

- Maximize $z = -x_1 - x_2 - x_3$
 subject to $x_1 - x_2 + 2x_3 = 2$,
 $-x_1 + 2x_2 - x_3 = 1$,
 $x_1, x_2, x_3 \geq 0$.

[Ans. $x_1 = 0$, $x_2 = \frac{4}{3}$, $x_3 = \frac{5}{3}$; $z_{max} = -3$.]

- 18 Use Charnes Big-M method to

- (i) Maximize $z = 3x_1 - x_2$, ($x_1, x_2 \geq 0$)
 subject to $2x_1 + x_2 \geq 2$,
 $x_1 + 3x_2 \leq 3$,
 $x_2 \leq 4$.

[Ans. $x_1 = 3$, $x_2 = 0$; $z_{max} = 9$.]

[Visva-Bharati, 1982]

- (ii) Maximize $z = 2x_1 - 3x_2$, $x_1, x_2 \geq 0$
 subject to $-x_1 + x_2 \geq -2$,
 $5x_1 + 4x_2 \leq 46$,
 $7x_1 + 2x_2 \geq 32$.

[Ans. $x_1 = 4$, $x_2 = 2$; $z_{max} = 2$.]

[Calcutta Hons., 1987]

19 Use Charnes M -method to solve the L. P. P.

- (i) Maximize $z = x_1 + 5x_2$
 subject to $3x_1 + 4x_2 \leq 6$ and $x_1 + 3x_2 \geq 3$,
 where $x_1, x_2 \geq 0$.

[Calcutta Hons., 1980]

[Ans. $x_1 = 0, x_2 = \frac{3}{2}; z_{\max} = \frac{15}{2}$.]

- (ii) Minimize $z = 4x_1 + 3x_2$
 subject to $x_1 + 2x_2 \geq 8$ and $3x_1 + 2x_2 \geq 12$,
 where $x_1, x_2 \geq 0$.

[Ans. $x_1 = 2, x_2 = 3; z_{\min} = 17$.]

20 Use Charnes Big M -method (method of penalty) to

- Maximize $z = 5x_1 - 2x_2 + 3x_3$
 subject to $2x_1 + 2x_2 - x_3 \geq 2$,
 $3x_1 - 4x_2 \leq 3$,
 $x_2 + 3x_3 \leq 5$,
 $x_1, x_2, x_3 \geq 0$.

[Kalyani Hons., 1982]

[Ans. $x_1 = \frac{23}{3}, x_2 = 5, x_3 = 0; z_{\max} = \frac{85}{3}$.]

21 Solve the following L. P. P. :

- Maximize $z = 2x_1 + x_2 + 3x_3$
 subject to $x_1 + x_2 + 2x_3 \leq 5$,
 $2x_1 + 3x_2 + 4x_3 = 12$,
 $x_1, x_2, x_3 \geq 0$.

[Jadavpur M. Sc., 1992]

[Ans. $x_1 = 3, x_2 = 2, x_3 = 0; z_{\max} = 8$.]

22 Find the minimum value of

- $2x_1 + 4x_2 + x_3$
 subject to $x_1 + 2x_2 - x_3 \leq 5$,
 $2x_1 - x_2 + 2x_3 = 2$,
 $-x_1 + 2x_2 + 2x_3 \geq 1$,
 $x_1, x_2, x_3 \geq 0$. [Kalyani Hons., 1981; Visva-Bharati Hons., 1988]

[Ans. 1.]

23 Find $x_j \geq 0, (j = 1, 2, 3, 4)$

- subject to $-2x_1 + x_2 + 3x_3 + x_4 = 2$,
 $-5x_1 + 2x_2 + 3x_4 = 5$,
 $7x_1 - 4x_2 + x_3 + 4x_4 \leq 1$,
 $3x_1 + 2x_2 + 5x_3 + 6x_4 \leq 10$,

which will minimize the function

$$-10x_1 + x_2 + 9x_3 + 8x_4.$$

[Ans. $x_1 = \frac{1}{7}, x_2 = \frac{8}{7}, x_3 = 0, x_4 = \frac{8}{7}$; Min. value = $\frac{62}{7}$.]

24 Show that the following two linear programming problems have no feasible solutions :

- (i) Maximize $z = 3x_1 + 2x_2$
 subject to $2x_1 + x_2 \leq 2$,
 $3x_1 + 4x_2 \geq 12$,
 $x_1, x_2 \geq 0$.

- (ii) Maximize $z = x_1 + 4x_2 + 3x_3$
 subject to $2x_1 - x_2 + 5x_3 = 40$,
 $x_1 + 2x_2 - 3x_3 \geq 22$,
 $3x_1 + x_2 + 2x_3 = 30$,
 $x_1, x_2, x_3 \geq 0$.

Check the result (i) graphically.

25 Show that the following two linear programming problems have unbounded solutions :

$$\begin{aligned} \text{(i) Minimize } z &= 3x - 2y \\ \text{subject to } x - y &\leq 1, \\ 3x - 2y &\leq 6, \\ x, y &\geq 0. \end{aligned}$$

$$\begin{aligned} \text{(ii) Maximize } z &= 5x_1 + 2x_2 + 2x_3 \\ \text{subject to } 3x_1 - 2x_2 - 2x_3 &= -8, \\ 3x_1 - 4x_2 - x_3 &= -7, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

Check the result (i) graphically.

26 Show that, in the linear programming problem

$$\begin{aligned} \text{Maximize } z &= 3x_1 + 4x_2 + 2x_3 \\ \text{subject to } x_1 - 2x_2 + 4x_3 &\leq 36, \\ 2x_1 + 3x_2 - 5x_3 &\leq 40, \\ 3x_1 + 2x_2 + x_3 &\leq 28, \\ x_1, x_2, x_3 &\geq 0, \end{aligned}$$

one of the resources need not be fully allocated to achieve the maximum value of z which is 56.

27 (a) Use Big-M method to maximize $z = 6x_1 + 4x_2$ subject to the constraints

$$\begin{aligned} 2x_1 + 3x_2 &\leq 30, \\ 3x_1 + 2x_2 &\leq 24, \\ x_1 + x_2 &\geq 3, \\ x_1, x_2 &\geq 0. \end{aligned}$$

Show that the solution is not unique. Find two solutions.

[Ans. $x_1 = 8, x_2 = 0; x_1 = \frac{12}{5}, x_2 = \frac{42}{5}; z_{\max} = 48.$]

(b) Find four alternative optimal basic solutions of the problem

$$\begin{aligned} \text{Maximize } z &= x_1 + 2x_2 + 3x_3 \\ \text{subject to } x_1 + 2x_2 + 3x_3 &\leq 10, \\ x_1 + x_2 &\leq 5, \\ x_1 &\leq 1, \\ x_1, x_2, x_3 &\geq 0. \end{aligned}$$

[Ans. $(0, 0, \frac{10}{3}), (0, 5, 0), (1, 4, \frac{1}{3}), (1, 0, 3).$]

28 Show that the problem

$$\begin{aligned} \text{Minimize } z &= -2x_1 + x_2 - 3x_3 \\ \text{subject to } x_1 - x_2 + 5x_3 &\leq 15, \\ -2x_1 + x_2 - 3x_3 &\geq 40, \\ x_1, x_2, x_3 &\geq 0 \end{aligned}$$

admits of alternative non-basic solutions.

29 A company produces two types of leather belt, say type A and B. Belt A is of a superior quality and belt B is of an inferior quality. Profits on the two types of belts are 40 paise and 30 paise per belt respectively. Each belt of type A requires twice as much time as required by a belt of type B. If all belts were of type B, the company could produce 1000 belts per day. But the supply of leather is sufficient only for 800 belts per day. Belt A requires a fancy buckle and only 400 buckles are available for this per day. For belt of type B, only 700 buckles are available per day.

How should the company manufacture the two types of belt in order to have a maximum over all profit?

[Ans. 200 A type and 600 B type belts; Max. Profit is Rs. 260.]

[Delhi, 1975]